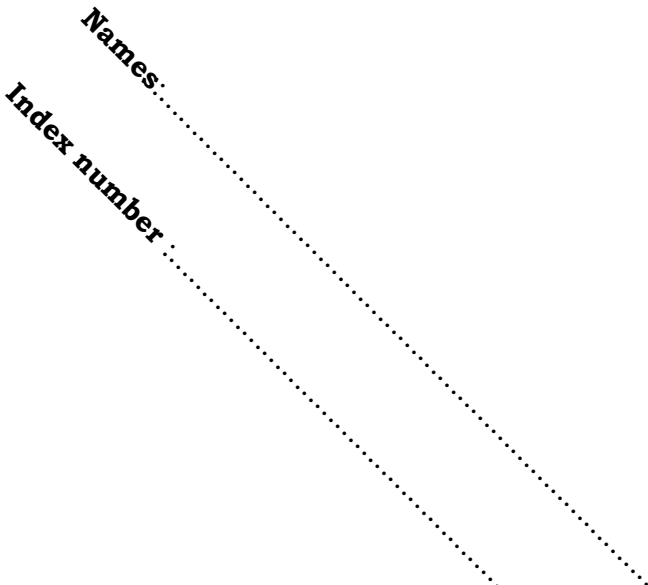




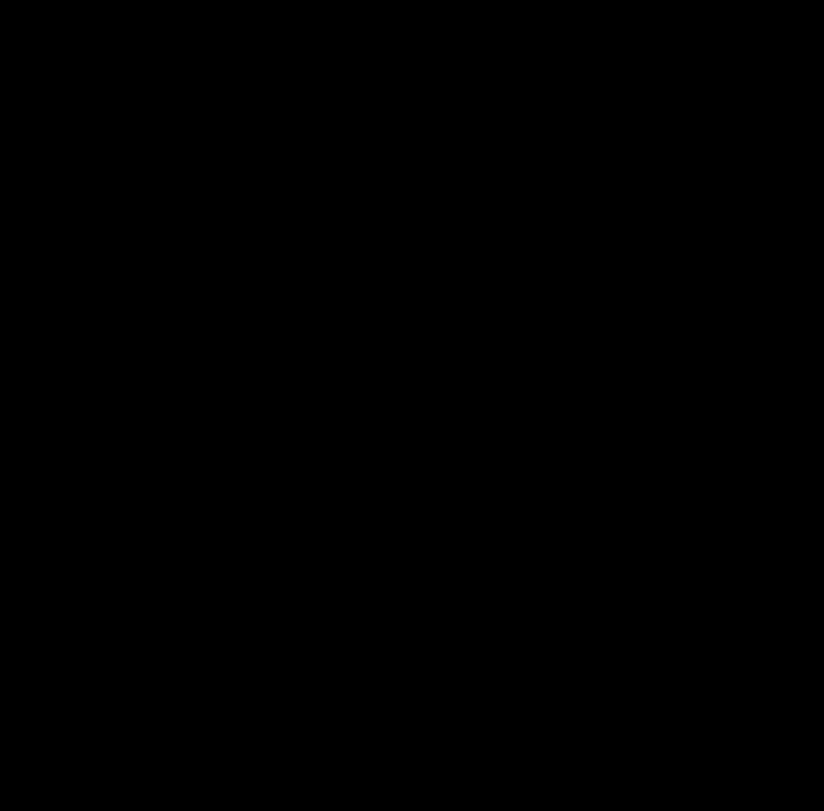
T2-148

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

TRADE (s) : SOFTWARE PROGRAMMING AND EMBEDDED SYSTEMS (SPE)

DURATION: 3 Hours

TOTAL MARKS...../100[illegible]



TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

TRADE: SOFTWARE PROGRAMMING AND EMBEDDED SYSTEMS (SPE)

**SUBJECT: RESTFUL SERVICES AND WEB/WEB3 APPLICATION
DEVELOPMENT**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

1. This Exam paper is composed of **three sections (A, B and C)**. Follow the instructions given below and answer the indicated questions for a total of **100 marks**.

Section **A**: Fourteen (14) questions which are **Compulsory** **55 marks**

Section **B**: Among the five (5) questions, attempt any three (3) **30 marks**

Section **C**: Among the two (2) questions, attempt anyone (1) **15 marks**

2. Fill in your actual **names** and **Index number** on the provided space (Cover).

3. Do not remove or tear any page or part of this Questions and Answers Booklet.

4. Answer in the language in which the Examination is set.

5. In section **A**, answer questions in the provided space. In case of multiple-choice questions, just **circle** the letter corresponding to the correct option.

6. In section **B** and **C**, answer questions in the provided space after section C questions.

7. **Allowed materials:**

- Blue or Black pen.
- Geometrical Instruments
- Silent non-programmable calculators

SECTION A: Attempt all questions**(55 marks)**

- 01.** Fill in the table below the software primarily associated with each graphic type. **(4marks)**

Graphic Type	Software
Vector Graphics	
Raster Graphics	

- 02.** Circle two (2) traits illustrating why Figma application is preferred over Adobe Photoshop for creating web-based software systems mockups **(4marks)**

- a) Collaboration
- b) Vector-based editing
- c) Photo retouching tools
- d) Components and libraries
- e) Interactive prototyping

- 03.** You have joined a software development team tasked with designing a student records management system for a university. As junior database designer, you are expected to assist in preparing the database schema and configure the DBMS environment. **(4marks)**

Question.

Which of the following items should you be able to recall and use during the interpretation of the database design and the initial DBMS setup?

(circle all that apply)

- a) Entity –relationship Diagram(ERD)
- b) Foreign Key Constraints.
- c) Sematic Versioning
- d) CREATE DATABASE Syntax with IF NOT EXISTS clause
- e) Grant command for assigning privileges
- f) Swagger for defining entity sets
- g) Table normalization using mathematical relations.

- 04.** Circle the two correct Assertion-reason(**A-R**) pairs regarding SQL database and table definition syntaxes: **(4Marks)**

- a) **(A)** the DROP DATABASE command removes the entire database including all tables.

(R) This is because SQL does not allow selective deletion of individual objects when using DROP DATABASE.

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- b) **(A)**The CHECK constraint validates column data against a specified condition
(B)It ensures that tall values in a column meet a defined logical expression.
- c) **(A)** The EXPORT operation can only be done using a GUI-based import/export wizard.
(R) Because SQL does not support command-line tools for exporting data.
- d) **(A)** The ALTER TABLE statement can be used to add new column to an existing table.
(R) SQL permits structural changes to tables after creation using the ALTER command.
- e) **(A)**Temporary tables can be accessed by all sessions in a multi-user environment.
(R) Because they are stored in the database schema just like permission tables.

- 05.** You are a junior backend developer joining a team that is building a Node.js-based RESTFUL API for a task management system. The team is following TEs-driven Development (TDD) practise, and you have been asked to write test before implementing new features. You are also expected to use express for routing and mongoose for dataset operations with Mongo DB. **(5Marks)**

Question.

Based on your knowledge of Node.js fundamentals and Test-driven Development, which of the following are relevant tools, concept or modules you would need to recall and use in this project? (Circle all that apply)

- a) Express.js for creating API routes
- b) Mongoose for connecting to and querying Mongo DB
- c) Swagger-UI for validating POST requests.
- d) Mocha or jest for writing unit tests
- e) Semantic versioning for documenting endpoint behaviour
- f) Nodemon for auto-restarting their server during development
- g) Async/Await for handling asynchronous database queries

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- 06.** You are working as a frontend developer on a decentralized finance (DeFi) project. Your responsibilities include building using React, connecting the frontend to a deployed smart contract and preparing yeapp for production deployment. You need to recall the tools and step used in this development stack. **(3Marks)**

Question.

Which of the following actions or tools should you remember and apply during the development and deployment of the web3-based interface from UI/UX design.

- a) Use React to implement user interfaces from UI/UX design.
- b) Use metamask to connect the frontend to a smart contract wallet.
- c) Use selenium to deploy production builds.
- d) Install web3.js to interact with smart contracts.
- e) Use firebase functions to create smart contracts.
- f) Configure environment variable for production deployment.

- 07. Assertion:** Proper use of HTML Structure and CSS styling directly influences a websites usability and user experience. **(4marks)**

Which of the following statements best explain that assertion? (Circle two correct answers).

Question:

- a) Block and elements help structure content logically, enhancing readability and accessibility
- b) Inline CSS is referred to over external CSS to keep the design consistent across multiple pages.
- c) CSS flexbox and responsive layouts improve how a site adapts to different devices and screen sizes.
- d) Using excessive SVG graphics and animations always improve ergonomics and visual comfort.
- e) HTML tables should be used for layout design instead of CSS to ensure uniformity.

- 08.** You are tasked with redesigning the homepage of a community health portal. The exiting page uses outdated HTML practices, lacks responsive layout and is hard to navigate on mobile devices. Your goal is to improve structure, styling and user experience based on good HTML and CSS practices and ergonomic guidelines. **(4marks)**

Which of the following actions would best improve the structure and usability of the web page? (circle two correct answers).

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- a) Replace <div> layout blocks with <table>tags to better control page alignment.
- b) Use semantic elements like <section>, <article> and <nav> to organize content meaningfully
- c) Apply inline styles to each to ensure immediately usual updates during development
- d) Implement CSS flexbox or grid to make the layout responsive across devices.
- e) Use bright saturated background colors and many fonts to make the interface look vibrant.

09. Assertion: Understanding asynchronous behavior and modular structure in Node.js is essential when building scalable and maintainable RESTFULL APIs with Express. **(3Marks)**

Which TWO of the following explanations best support this assertion?

- a) Using require() to import modules allows developers to organize functionality into reusable components, reducing code duplication.
- b) Writing all logical onside a single Express route handler ensures better performance and simpler code management.
- c) Asynchronous patterns like promises an async/await help in managing non-blocking operations such as database calls.
- d) Synchronous JavaScript functions ate preferred in express apps because they make debugging easier.
- e) Express automatically converts all synchronous code into asynchronous code during request handling.

10. Match the ReactJS concepts with its correct explanation. **(4Marks)**

- | | |
|-------------------|--|
| a) JSX in React | 1. Techniques for passing data from parent to child components. |
| b) Props | 2. Mechanism for listening to and responding to user interaction |
| c) Event handlers | 3. JavaScript Syntax extension that allows writing HTML in JS files. |
| d) React State | 4. Data storage for managing dynamic components-specific information |

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- 11.** You are working as a database analyst for an online retail company. **(4Marks)**
Your manager asked you to identify all customer orders placed during January 2024, stored in descending order of total purchase amount. You need to write a query to extract this information from the orders table which contains the columns order_id, customer_id, order_date and total_amount.

Question

Which of the following SQL queries would correctly retrieve the required data? (circle the best option (s))

- a) `SELECT *
FROM orders
WHERE order_date BETWEEN '2024-01-01' AND '20234-01-31'
ORDER BY total_amount DESC;`
- b) `SELECT *
FROM orders
WHERE MONTH(order_date) =1 AND YEAR(order_date) =2024
ORDER BY total_amount DESC;`
- c) `SELECT *
FROM orders
WHERE order_date >= '2024-01-01' AND order_date<= '2024-01-31'
ORDER BY total_amount DESC;`
- d) `SELECT *
FROM orders
WHERE YEAR(order_date) =2024 AND MONTH(order_date) =1
ORDER BY total_amount DESC;`

- 12.** You are developing a PHP-based inventory management system for a retail store. You want to create reusable components using OOP principles and ensure that each product object can manage its own properties securely. **(4Marks)**

Question

Based on this requirement, which of the following technique would you apply to implement this functionality effectively?(circle the right option)

- a) Use private visibility and getter/setter methods to encapsulate product data
- b) Apply final keyword to allow product class to be extended by future modules.
- c) Use public instance variables to exposure details directly.
- d) Define a class and create object instances to manage each product.

- 13.** You are a developer working on a blockchain application for a logistics company. Each node in the network must automatically discover peers to share data and you need to manage unconfirmed transactions before they are mined into blocks. **(4Marks)**

Question

Based on this scenario, which of the following tools or techniques would you apply to support peer discovery and transaction management? (Circle two correct answers)

- a) Use discovery–swarm to enable nodes to find and connect to peer.
- b) Use elliptic to implement consensus algorithms like proof of stake.
- c) Implement transaction pools to manage not yet validated wallet transactions.
- d) Apply LevelDB to persist and retrieve blockchain data efficiently.
- e) Store wallet balances in local variables to simplify mining calculations.

- 14.** You are designing a decentralized application (dApp) that must process transactions quickly and minimize energy consumption. Based on your understanding of consensus mechanism, **(4Marks)**

Question:

Which of the following option would best meet these requirements? (Circle two correct answers)

- a) Proof of Stake (PoS), as it allows faster block validation with lower energy.
- b) Proof of Work (PoW), it guarantees decentralization with minimal energy usage.
- c) Proof of stake (PoS), since it selects validators based on stake instead of energy-intensive mining.
- d) Proof of Stake (PoS), as it accelerates transaction speed by prioritizing weather miners using with less energy.

Section B: Attempt any three (3) questions**(30 marks)**

- 15.** Consider two tables in a MySQL database for a local e-commerce platform in Rwanda: customers (customer_id, name, region) and orders (order_id, customer_id, order_date, total_amount). Analyze the differences in the result sets you would obtain when performing an INNER JOIN, a LEFT JOIN (with customers as the left table), and a RIGHT JOIN (with customers as the right table) to combine these tables based on customer_id. Explain which join type would be most suitable if you need to retrieve all customers and their corresponding orders, even if a customer has not placed any orders yet. **(10marks)**
- 16.** Analyze below code snippet to answer related questions: **(10marks)**
- ```
1) Function findSum(... myArgs){
2) return myArgs.reduce((acc, curr)=> {
3) if (typeof curr === 'number') {
4) return acc+curr;
5) }else if (typeof curr === 'string'){
6) return acc+ curr.length;
7) }
8) return acc;
9) },0);
10) }
11) Console.log (findSum(1, 2, 'Hello', 3, 'world'));
12) Console.log(findSum(4, null, 'test', true,5));
```
- i) What is the output on line 11?  
ii) What is the output on line 12?
- 17.** Examine below code snippet that depicts a customized node.js middleware, to answer related questions after assuming that all dependencies are installed **(10marks)**
- ```
1. /*simple hello world in node.js*/
2. Const myMiddleware = (req, res, next) => {
3.   Console.log('Request received:', req.method, req.url);
4.   //Perform some task
5.   req.message= 'Hello from middleware';
6.   //call the next middleware function
7.   next();
8. };
9. //Example Express application
```

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```
10. const express = require('express');
11. const app = express ()
12. //use the middleware
13. App.use (myMiddleware)
14. //Example route handler
15. app.get('/', (req, res) => {
16.   console.log('Message from middleware:', req.message);
17.   res.send('Hello, world!');
18. });
19. //start the server
20. app.listen(3000, () => {
21.   console.log(server listening on port 3000);
22. });
```

- i) What will be the browser displayed from this URL <http://localhost:3000/> ?
- ii) What will be the output on line 16?

- 18.** Observe below table and snippet code and rewrite PHP missed code to **(10marks)** display data as shown in the table from database

Student_no	Student_name	Subject	Marks
1	Kalisa	Database	90
2	Shyaka	English	70

```
<?php
include 'connection.php';
$sql = 'SELECT * FROM marks';
$result = $conn->query($sql);
?>
<table class="table">
<thead>
<tr>
<th> Student_no </th>
<th> Student_name</th>
<th> Subject </th>
<th> Marks </th>
</tr>
</thead>
<tbody>
```

19. Evaluate the following code snippet:

(10marks)

```
1. function greet(name) {  
2.   return "Hello," + name;  
3. }  
4.
```

- i) If the function call is greet("Alice");
What is the output?
- ii) Ignore double quotes (") in the output to select what will be the output of the following code?

```
5. console.log(greet(null));  
6. console.log(greet(undefined));  
7. console.log(greet(42));  
8. console.log(greet(" "));
```

Section C: Attempt only one (1) question

(15 marks)

- 20.** You are a junior developer at a company that is building a “Learning companion App” for primary school students. One of the features in the app is a basic arithmetic module that helps students practice number series and calculations. **(15marks)**

Your team leader has asked you to implement features where the system calculate the sum of all natural numbers from 1 to 10 using object-oriented programming in PHP. This functionality will later be extended to allow flexible ranges, so it must be encapsulated in class.

Your task is to write PHP class named SeriesCalculator with a method called SumSeries() that return sum of the natural numbers from 1 to 10.

- 21.** You are working as a junior developer for a startup building lightweight educational tool that can run on local networks in school. Your team leader asks you to demonstrate a simple web server that displays a “Hello World message” in a browser. **(15marks)**

To prototype this feature You are instructed to use PHP’s built in server and ensure it listens on port 30000 so teachers can access it via <http://localhost:3000>. Create a server to display Hello world on a browser and runs on port 3000.

Create a PHP file that outputs “Hello World” to the browser and set up a local server to run it on port 3000 using PHP built-in development server.

END

RESERVED SPACE FOR SECTION B AND C ANSWERS

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